

Curriculum Vitae

Diogo Afonso Pedro Nunes

(alias, Diogo A.P. Nunes)

NLP/ML Researcher & Engineer @ INESC-ID/HLT
PhD with Distinction and Honor @ IST

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SUMMARY

Natural Language Processing (NLP) / Machine Learning (ML) researcher & engineer with 5+ years of experience applying NLP/ML to real-world problems in healthcare. International research experience leveraging and fine-tuning LLMs. Experience in data science and data collection at scale. International NLP competition winner for depression symptoms detection in text. Award-winning peer-reviewed studies. Publications on high-impact Artificial Intelligence (AI) conferences. PhD in Computer Science & Engineering.

SKILLS & TECHNOLOGIES

ML/NLP: PyTorch, Hugging Face Transformers, scikit-learn, TensorFlow, vLLM, Ollama, Git, LangChain, LangGraph, NLTK, SpaCy, Pandas, NumPy

Programming Languages: Python, C#, C, C++

Other: AWS, Data Science, Clinical NLP, Predictive Modeling, Scientific Writing, Oral Communication

EXPERIENCE

Visiting Researcher

Northwestern University – Chicago, United States

March 2024 – June 2024

- Fine-tuned LLM for domain adaptation with web scraped data. Predictive modeling of patient clinical state, based on LLM encoding, SVM and Decision Tree models. Text mining and clustering coupled with LLM prompting for interpretability.

Researcher

INESC-ID, Human Language Technologies Lab – Lisbon, Portugal

May 2019 – Present

- LLM fine-tuning, chain-of-thought prompting, few-shot prompting, RAG, and Information Retrieval techniques for symptom depression detection in text. Data analysis, feature engineering, and predictive modeling of patient clinical state. Web scraping at scale, topic modeling, and graph analysis of social media text.

IT Infrastructure Solutions

INESC-ID – Lisbon, Portugal

January 2022 – December 2024

- Design and implementation of internal IT services, including intranet and related databases. Implementation with .NET and C#. Docker containerization. Production-grade deployment.

Junior Internships (Data Scientist & Software Developer)

CEiiA – Porto, Portugal (July 2019 – August 2019)

EDP Innovation – Lisbon, Portugal (July 2018 – August 2018)

Building Global Innovators – Lisbon, Portugal (May 2017 – October 2017)

INVITED TALKS & LECTURES (selected)

- **Keynote Speaker @ Aarhus University**, Sep 2025: “Computational Models of the Language of Pain: Exploring Patient Narratives with Natural Language Processing”
- **Invited Lecturer @ IST, University of Lisbon (BEST Summer Course)**, Jul 2025: “How to Save a Life: Redefining Healthcare with AI-Wearables”
- **Invited Lecturer @ ISCTE, University of Lisbon (Computational Language Processing)**, Jun 2025: “Computational assessment of chronic pain based on patient language of pain”
- **Teaching Assistant @ IST, University of Lisbon (Compilers)**, February 2020 – July 2020

EDUCATION

- **PhD in Computer Science and Engineering (2021 – 2025)**
Instituto Superior Técnico, University of Lisbon – Lisbon, Portugal
Pass with Distinction and Honor
- **MSc in Computer Science and Engineering (2018 – 2021)**
Instituto Superior Técnico, University of Lisbon – Lisbon, Portugal
ERASMUS Exchange – Paris, France @ EFREI Paris
- **BSc in Computer Science and Engineering (2015 – 2018)**
Instituto Superior Técnico, University of Lisbon – Lisbon, Portugal

AWARDS & RECOGNITION (selected)

- **Best Clinical Research Paper**. Portuguese Association for the Study of Pain
- **1st Place**. eRisk 2025 challenge: depression symptom detection, CLEF
- **Finalist**. “3 Minute Thesis” Competition (2023), University of Lisbon

PUBLICATIONS (selected)

- Nunes et al., *Language as a Predictor of Multiple Clinical Chronic Pain Assessments*. Accepted @ ECAI Oct 2025, Italy.
- Nunes et al., *INESC-ID @ eRisk 2025: Exploring Fine-Tuned, Similarity-Based, and Prompt-Based Approaches to Depression Symptom Identification*. Accepted @ CLEF Sep 2025, Spain. Preprint: [10.48550/arXiv.2506.02924](https://arxiv.org/abs/10.48550/arXiv.2506.02924)
- Nunes et al., *Advancing the prediction and understanding of placebo responses in chronic back pain using large language models*. Under review. 2025. Preprint: [10.1101/2025.01.21.25320888](https://arxiv.org/abs/10.1101/2025.01.21.25320888)
- Nunes et al., *Modeling chronic pain experiences from online reports using the Reddit Reports of Chronic Pain dataset*. Information, MDPI. 2023. DOI: [10.3390/info14040237](https://doi.org/10.3390/info14040237)